Spherical DRA in CST: Build-from-Scratch Tutorial

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Overview

You will build a 10 mm alumina sphere on a 0.5 mm Rogers RO3003 substrate. Two offset 1 mm microstrip lines feed it, and the sphere resonates at about 9.96 GHz (magnetic-dipole mode) and about 13.5 GHz (electric-dipole mode) inside a 9–14 GHz sweep.

The sphere sits in a shallow hole drilled through the substrate and touches the ground plane at one point. Each feed line stops 1 mm short of the hole. Energy couples into the sphere from line 1 through the fringing fields and out to line 2, so the S21 response peaks at the sphere's resonances.

As modelled, this is a **two-port coupled-resonator structure**: it measures S11 and S21 and has no far-field monitors. To use it as a radiating DRA, follow the last step of the build and the pitfalls section: open the top boundary, add far-field monitors and terminate port 2 (or remove it).

```
flowchart LR
    A[New project<br/>units + solver] --> B[Parameters]
    B --> C[Materials]
    C --> D[Substrate + sphere<br/>+ seating hole]
    D --> E[Microstrip feeds<br/>mirror x2]
    E --> F[Boundaries<br/>+ ports]
    F --> G[Solver +<br/>monitors]
    G --> H[Run +<br/>post-process]
```

The build follows the order of the original ModelHistory: each step below matches one or more entries in the History List.

Part 0A — CST for absolute beginners

CST Studio Suite is a 3D electromagnetic simulator. You draw a structure, assign a real material to every part, tell CST where the signal goes in and out (ports) and what surrounds the model (boundaries), and it solves Maxwell's equations to give S-

parameters and field plots. Parts 0A–0C teach just enough CST to follow Steps 1–10. You don't need any prior CST experience.

What you need before starting

- **CST Studio Suite installed with a working licence.** Your college lab or department usually provides this. Any version from 2021 onward works; your model was made in 2021 and last saved in 2025. Menu names can differ slightly between versions, but every button in this guide exists in all of them.
- **A Windows PC with 16 GB RAM or more.** This model is small, so a typical lab PC runs it in a few minutes.
- **A folder for your work**, for example `D:\BTP\CST\`. Avoid OneDrive or other synced folders, because CST writes many result files.

The whole job in one sentence

Open CST → pick a template → define parameters → load materials → draw 5 shapes → set boundaries → add 2 ports → set up the solver → click Start → read the S-parameter graph.

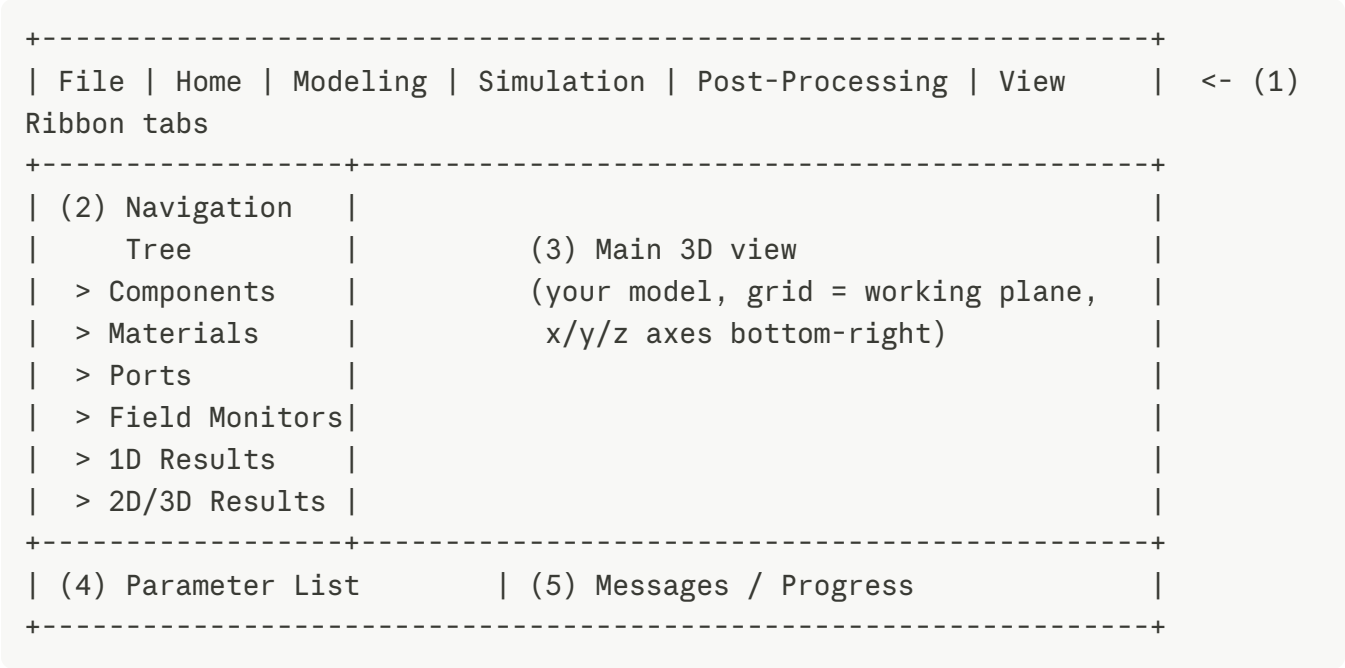
Key terms you will see

Term	Meaning in plain words
MWS (Microwave Studio)	The part of CST for antennas, filters and resonators. This is the one you use.
Template	A starting preset that fills in units, solver and boundaries for a type of device
Solid / Shape	One 3D object (brick, sphere, cylinder). Each has a name, a component and a material.
Component	A folder that groups solids in the navigation tree (here, `component1`)
Parameter	A named number such as `r_sphere = 5`. Shapes use names, so changing one value updates the model.
Material	Electrical properties: εr (permittivity), tanδ (loss), σ (conductivity)

Term	Meaning in plain words
PEC	Perfect electric conductor: ideal metal with no loss
Boundary	What sits at the six walls of the simulation box: open (free space) or electric (a metal wall)
Waveguide port	A rectangle where CST injects and measures the signal
S-parameters	S11 = how much signal reflects back to port 1; S21 = how much gets through to port 2. Measured in dB.
Solver	The maths engine. You'll use the **Frequency Domain** solver.
Mesh	The small tetrahedra CST splits the model into to solve it
Monitor	An instruction to save the full 3D field at one frequency so you can view it afterwards
History List	CST's record of every step you took. It can replay the whole model, and it's what your `model.mod` file is.

Part 0B — A tour of the CST window

Once a project is open, the CST window has five areas. Learn them once and every step below makes sense.



#	Area	What you use it for
1	**Ribbon tabs**	Every command. **Home**: units, parameters, history, macros. **Modeling**: shapes, materials, booleans, transforms, picks. **Simulation**: frequency, boundaries, ports, monitors, solver setup, Start. **Post-Processing**: result templates and exports. **View**: show or hide windows, wireframe, cut planes.
2	**Navigation Tree** (left)	Lists everything in the model. Click a solid to select it, double-click to edit it, right-click for options. After a run, results appear under *1D Results* (graphs) and *2D/3D Results* (field plots).
3	**Main view** (centre)	The 3D model. The grid is the *working plane* (z = 0 by default), and the small x/y/z arrows show orientation.
4	**Parameter List** (bottom)	All parameters: name, expression, value. You can edit them here at any time.
5	**Messages / Progress** (bottom)	Warnings, errors and solver progress. Read it when something looks wrong.

If you close a window by mistake, go to **View → Window** (or the *Windows* dropdown on the View tab) and tick it again.

The Navigation Tree folders you'll touch

- **Components → component1**: holds `solid1` (substrate), `solid2` (sphere), `solid4` and `solid4_1` (feed lines).
- **Materials**: shows the materials you loaded, with their colours.
- **Ports**: port 1 and port 2.
- **Field Monitors**: the E, H and power monitors.
- **1D Results → S-Parameters**: S1,1 and S2,1 graphs after the run.
- **2D/3D Results → E-Field / H-Field**: field plots after the run.

Part 0C — Six core skills to practise first

Spend 15 minutes on these in a throw-away project before building the real model. Every step later on is a combination of them.

Skill 1 — Move the 3D view

- **Rotate:** hold the left mouse button and drag.
- **Zoom:** scroll the mouse wheel.
- **Pan:** drag with the middle (wheel) button, or use the Pan tool on the **View** tab.
- **Fit the whole model to the screen:** press the **Spacebar** (Reset View).
- **See through solids: View → Wireframe**. Use **View → Cutting Plane** to slice the model and look inside, for example to check the sphere sitting in its hole.

If the mouse behaves differently on your version, the Rotate, Pan and Zoom buttons on the **View** tab always work.

Skill 2 — Draw a shape by typing exact values (the Esc trick)

This is the most important skill. Never draw with the mouse; always type the coordinates.

1. Click a shape button, for example **Modeling → Shapes → Brick**.
2. The status bar asks you to pick points. Ignore that and **press Esc**. A dialog box opens.
3. Fill in Name, Component, Material and the X/Y/Z ranges, using parameter names such as `l_sub/2` instead of numbers.

4. Click **Preview** to check it, then **OK**.

If you type a parameter name that doesn't exist yet, CST asks for its value and creates it for you. That's a handy way to add parameters on the fly.

Skill 3 — Handle the "Shape Intersection" dialog

When a new solid overlaps an existing one, CST asks what to do:

Option	What to do in this project
None (keep both as they are)	Choose this when the sphere overlaps the substrate and when the vacuum cylinder overlaps it; Step 5c then removes the overlap cleanly.
Insert / Trim / Add	These options cut or merge the two solids automatically. Don't use them here, because they would give different solid names from the original model.
Cancel	Use this if you drew the wrong shape.

The feed lines sit on top of the substrate and touch it without overlapping, so this dialog won't appear for them.

Skill 4 — Pick a face (needed for ports)

1. **Modeling → Picks → Pick Face**, or press **F**.
2. Hover over the model. Faces highlight as you move.
3. Double-click the face you want. It turns red or orange and stays selected.
4. Now open the command that needs it (for example **Simulation → Waveguide Port**).
5. To clear a wrong pick: **Modeling → Picks → Clear Picks**.

Zoom in (Skill 1) so you click the tiny 1 mm × 0.018 mm end face of the line, not the substrate edge next to it.

Skill 5 — Edit or undo something

- **Undo the last step:** Ctrl+Z.

- **Change an old step,** such as a wrong dimension: **Home → History List**, select the line (for example *define brick: component1:solid1*), click **Edit**, fix the value, click **OK**, then **Update**. CST replays everything after that step.
- **Change a dimension that uses a parameter:** just edit its value in the Parameter List, then click **Home → Update** (or answer *Yes* when CST asks to rebuild).
- **Delete a solid:** select it in the Navigation Tree and press Delete.

Skill 6 — Save correctly

- Press **Ctrl+S** often. CST saves `sphere_DR.cst` and a folder `sphere_DR\` next to it.
- **Always keep the .cst file and its folder together.** The folder holds the mesh and the results. If you copy only the .cst file, you lose your results.
- To share or back up a project, use the **Archive** option in the File menu (it packs the project into a single file), or copy both items.
- Your `model.mod` and `ModelHistory.json` files live inside that folder, under `Model\3D\`. That's where the files you sent me came from.

Components and materials

The model has five solids, two ports and one ground plane. The ground is not a solid: the Zmin electric boundary supplies it.

Component	CST shape / name	Material	εr	tanδ / σ	Role
Substrate	Brick `solid1`	Rogers RO3003 (loss free)	3.0	0	Carries the microstrip lines and holds the sphere
Seating hole	Cylinder `solid3`, subtracted from `solid1`	Vacuum	1.0	—	Pocket so the sphere can sit through the substrate onto the ground
Resonator	Sphere `solid2`	Alumina (99.5%) (lossy)	9.9	tanδ = 1e-4	The dielectric resonator itself
Feed line 1	Brick `solid4`	Copper (annealed)	—	σ = 5.8e7 S/m	Input microstrip line (port 1)
Feed line 2	Mirrored copy `solid4_1`	Copper (annealed)	—	σ = 5.8e7 S/m	Output microstrip line (port 2)
Ground plane	Zmin = electric boundary	PEC	—	—	Microstrip ground and the sphere's image plane

Component	CST shape / name	Material	εr	tanδ / σ	Role
Ports 1 and 2	Waveguide ports at x = ±25 mm	—	—	—	Excitation and S-parameter reference planes

All three materials ship in the CST material library, so you don't need to type them in by hand (see the materials step).

Parameters

Define these parameters before drawing anything, so every dimension can be swept later. All lengths are in mm and frequencies in GHz.

Name	Expression	Value	Meaning
`l_sub`	50	50	Substrate length (x)
`w_sub`	30	30	Substrate width (y)
`t_sub`	0.5	0.5	Substrate thickness (z)
`r_sphere`	5	5	Sphere radius. This is the main tuning knob for frequency.
`w_inset`	`(r_sphere^2-(r_sphere-t_sub)^2)^0.5`	2.1794	Hole radius, equal to where the sphere crosses the top of the substrate
`s_mic`	1	1	Gap between the end of each line and the hole edge (sets coupling)
`w_mic`	1	1	Microstrip width
`ss_mic`	3	3	Offset of each line from the y = 0 centreline (sets coupling to the mode)
`t_metal`	0.018	0.018	Copper thickness (18 µm, ½ oz)
`f_min`	9	9	Sweep start
`f_max`	14	14	Sweep end

Name	Expression	Value	Meaning
`s_sphere`	1	1	Unused in this model (leftover)
`x`	`8*r_sphere+3.5*s_sphere+s_mic`	44.5	Unused in this model (leftover, possibly from a multi-sphere version)
`r_wire`	0.15	0.15	Unused in this model (leftover)

You can skip `s_sphere`, `x` and `r_wire`. They don't affect this geometry.

Design theory (hand calculations)

The sphere radius and permittivity set the resonant frequencies. The feed geometry only sets how strongly each mode is excited.

1. Resonant frequency of the sphere

For a dielectric sphere of radius a and refractive index n = $\sqrt{\varepsilon_r}$, the magnetic-dipole mode (TE_{101}) resonates at roughly $n \cdot k_0 \cdot a \approx \pi$:

```
f_{TE101} \approx \frac{c}{2\,a\,\sqrt{\varepsilon_r}} = \frac{3\times10^{8}}
{2 \times 0.005 \times 3.146} \approx 9.5\ \text{GHz}
```

The electric-dipole mode (TM_{101}) resonates at roughly $n \cdot k_0 \cdot a \approx 4.49$:

```
f_{TM101} \approx \frac{4.49\,c}{2\pi\,a\,\sqrt{\varepsilon_r}} \approx 13.6\
\text{GHz}
```

These are rough estimates for an isolated sphere. The ground plane, the substrate and the feeds pull the results by a few percent, which matches the monitors at 9.957, 10.34, 13.45 and 13.6 GHz.

To design for a different frequency, scale the radius. Since $f \propto 1 / (a \cdot \sqrt{\varepsilon_r})$, halving `r_sphere` roughly doubles the frequency.

2. Seating-hole radius

The sphere centre sits at z = a, so its bottom touches the ground at z = 0. At the top of the substrate (z = t), its cross-section has radius:

```
w_{inset} = \sqrt{a^2 - (a - t)^2} = \sqrt{25 - 20.25} = 2.179\ \text{mm}
```

Cutting a cylinder of this radius through the substrate removes every region where the substrate would overlap the sphere.

3. Microstrip line width

The Hammerstad formulas give the line impedance for w/h ⩾ 1:

```
\varepsilon_{eff} = \frac{\varepsilon_r+1}{2} + \frac{\varepsilon_r-1}
{2}\left(1+\frac{12h}{w}\right)^{-1/2}
```

```
Z_0 = \frac{120\pi}{\sqrt{\varepsilon_{eff}}\left[\frac{w}{h} + 1.393 +
0.667\ln\left(\frac{w}{h}+1.444\right)\right]}
```

Width w (mm)	w/h	εeff	Z_0 (Ω)
1.00 (as in this model)	2.0	2.38	≈ 58
1.25	2.5	2.42	≈ 50

The original model uses 1 mm lines, which gives a mild mismatch against the 50 Ω port normalisation. Use `w_mic = 1.25` if you want true 50 Ω lines.

4. Coupling knobs

- `s_mic` (end gap): a smaller gap gives stronger coupling, a wider bandwidth and a lower loaded Q.
- `ss_mic` (lateral offset): moves each line end through the sphere's field pattern, which changes which modes are excited and how strongly.
- The two lines are offset in opposite directions (point symmetry), so modes that are odd across y = 0 can couple through.

Step 1 — Create the project (click by click)

Use the **Project Template wizard** to create a Microwave Studio project with mm/GHz units and the Frequency Domain solver, then save it to your own folder. The wizard has about six screens and takes about two minutes.

1a. Launch CST and open the wizard

1. From the Windows Start menu, open **CST Studio Suite <year>**. The start page appears, showing recent projects and a *Create Project* area.
2. Under *Create Project*, click **New Template**. If you are already inside a project, use **File → New and Recent → New Template** instead.

Don't click one of the big module tiles such as *Microwaves & RF/Optical* directly. That opens an empty project without the wizard's settings.

1b. Go through the wizard, screen by screen

Screen	What CST asks	What you choose	Why
1	Application area	**Microwaves & RF/Optical**	Selects Microwave Studio (MWS), the right module for resonators and antennas
2	Workflow	**Antennas → Planar (Patch, Slot, etc.)**. If your version lists a *Dielectric Resonator* workflow, that is fine too.	Planar templates suit microstrip feeds on a substrate
3	Solver	**Frequency Domain** (not Time Domain)	Best for small, high-Q resonators with curved surfaces
4	Units	Dimensions **mm**, Frequency **GHz**, Time **s**, Temperature **Kelvin** (leave the others at default)	Matches every number in this guide
5	Settings	Frequency Min **9**, Max **14**. Untick all monitors for now; you add them in Step 9.	Sets the 9–14 GHz sweep

Screen	What CST asks	What you choose	Why
6	Summary	Name the template, for example `DR_9-14GHz`, then click **Finish**	CST saves the template so you can reuse it for the next design

Click **Next** between screens. If a screen looks different in your version, pick the closest match. Steps 1c–1d and Step 7 correct any template setting that matters.

1c. Save immediately, in your own folder

1. **File → Save As**.
2. Browse to your work folder, for example `D:\BTP\CST\`.
3. Name it `sphere_DR` and click **Save**.
4. CST now creates `sphere_DR.cst` and a folder `sphere_DR\`. Keep them together (Part 0C, Skill 6).

1d. Check that the template did its job

Check	Where	Should show
Units	**Home → Units**	mm, GHz, s
Solver type	Solver icon on the **Home** tab, or **Simulation → Setup Solver**	Frequency Domain solver dialog
Frequency range	**Simulation → Frequency**	9 to 14 GHz. You'll replace these with `f_min` and `f_max` in Step 9.
Boundaries	**Simulation → Boundaries**	Anything for now; you set them in Step 7

If the solver is wrong: **Home → Setup Solver** dropdown (or the solver-type dropdown on the Simulation tab) → choose **Frequency Domain Solver**.

1e. Get the windows ready

- Make sure the **Parameter List** and **Messages** windows are visible at the bottom (Part 0B). Use **View → Window** if they're missing.

- Open **Home → History List** once to see where CST records your steps; it's empty or nearly empty now. Close it again.

You now have an empty, correctly configured project. Continue with Step 2 (parameters).

Step 2 — Define the parameters

Enter every parameter from the Parameters table before you draw.

1. Open the **Parameter List** window. If it's hidden, re-open it from the **View** tab (Window → Parameter List).
2. Click the empty *new parameter* row, type the name, press Enter, and type the expression.
3. Enter the base values first: `l_sub`, `w_sub`, `t_sub`, `r_sphere`, `s_mic`, `w_mic`, `ss_mic`, `t_metal`, `f_min` and `f_max`.
4. Then enter the derived one: `w_inset` = `(r_sphere^2-(r_sphere-t_sub)^2)^0.5`. The list should show 2.17945.

Tip: always type parameter names, never numbers, into the shape dialogs. Changing `r_sphere` later then rebuilds the sphere, the hole and the line positions automatically.

Step 3 — Load the materials

Import three materials from the CST library: **Modeling → Materials → Material Library**. Search for each name, then click **Load**.

Library name	εr	Loss	Used for
Rogers RO3003 (loss free)	3.0	tanδ = 0	Substrate
Alumina (99.5%) (lossy)	9.9	tanδ = 1e-4	Sphere
Copper (annealed)	—	Lossy metal, σ = 5.8e7 S/m	Feed lines

Vacuum is built in and needs no import.

If you prefer a realistic substrate, load **Rogers RO3003 (lossy)** instead (tanδ ≈ 0.001). Expect slightly more insertion loss and a lower Q.

Step 4 — Draw the substrate

The substrate is a brick centred on the origin whose bottom face is the ground plane (z = 0).

1. Create the component: **Modeling → Components → New Component**, and name it `component1`.
2. **Modeling → Shapes → Brick**, press **Esc** to open the dialog, and enter:

Field	Value
Name	`solid1`
Component	`component1`
Material	Rogers RO3003 (loss free)
Xmin / Xmax	`-l_sub/2` / `l_sub/2`
Ymin / Ymax	`-w_sub/2` / `w_sub/2`
Zmin / Zmax	`0` / `t_sub`

Step 5 — Add the sphere and the seating hole

Place the sphere so its bottom touches z = 0, then cut a hole in the substrate where the two overlap.

5a. Sphere: Modeling → Shapes → Sphere, then Esc.

Field	Value
Name	`solid2`
Material	Alumina (99.5%) (lossy)
Axis	z
Center radius	`r_sphere`
Top / Bottom radius	`0` / `0`
Center (x, y, z)	`0`, `0`, `r_sphere`

Field	Value
Segments	`0` (smooth true sphere)

CST may show a **Shape Intersection** dialog because the sphere overlaps the substrate. Choose **None** (keep both); the hole in 5b resolves the overlap.

Optional: raise the shape visualisation accuracy to 120 so the sphere renders smoothly. The history line is `Solid.ShapeVisualizationAccuracy2 "120"`. This setting is display-only.

5b. Hole cylinder: Modeling → Shapes → Cylinder, then Esc.

Field	Value
Name	`solid3`
Material	Vacuum
Axis	z
Outer / Inner radius	`w_inset` / `0`
Xcenter / Ycenter	`0` / `0`
Zmin / Zmax	`0` / `t_sub`

5c. Subtract: select `solid1` in the navigation tree, then **Modeling → Boolean → Subtract**, pick `solid3` and press Enter. The substrate now has a round hole of radius 2.18 mm.

Check: in a cut-plane view (y = 0), the sphere sits in the hole and touches the ground at its lowest point. A small vacuum ring between the sphere and the hole wall is expected.

Step 6 — Draw the microstrip feeds

Draw one line on the −x side, then mirror it twice to get the point-symmetric second line on the +x side.

6a. Line 1: Modeling → Shapes → Brick, then Esc.

Field	Value	Evaluates to
Name	`solid4`	—

Field	Value	Evaluates to
Material	Copper (annealed)	—
Xmin / Xmax	`-w_inset-s_mic` / `-l_sub/2`	−3.18 / −25
Ymin / Ymax	`-w_mic/2-ss_mic` / `w_mic/2-ss_mic`	−3.5 / −2.5
Zmin / Zmax	`t_sub` / `t_metal+t_sub`	0.5 / 0.518

The line runs from the substrate edge (x = −25) to 1 mm short of the hole, and sits 3 mm below the centreline.

6b. First mirror, which creates a copy: select `solid4`, then **Modeling → Transform → Mirror**.

- Mirror plane normal: (0, 1, 0), through the origin
- **Copy** ticked. This is the history's `MultipleObjects "True"`.
- Result: a new `solid4_1` at y = +2.5 to +3.5, still on the −x side

6c. Second mirror, which moves the copy: select `solid4_1`, then **Transform → Mirror**.

- Mirror plane normal: (1, 0, 0), through the origin
- **Copy** unticked
- Result: `solid4_1` moves to x = +3.18 to +25

You now have two lines that map onto each other under a 180° rotation about z: line 1 on the lower left, line 2 on the upper right, as in the reference screenshot.

Why the offset matters: a centred pair (`ss_mic` = 0) would couple to a different set of sphere modes. The ±3 mm offset is part of the design, so keep it unless you are deliberately studying coupling.

Step 7 — Set the boundaries

Open **Simulation → Boundaries**. The Zmin boundary must be electric, because it is the ground plane; the sides are open.

Face	Original model	Recommended for a radiating DRA
Xmin, Xmax	expanded open	expanded open
Ymin, Ymax	expanded open	expanded open

Face	Original model	Recommended for a radiating DRA
Zmin	electric (E_t = 0)	electric (ground plane)
Zmax	electric	**expanded open**
Symmetry planes (Symmetry tab)	none	none
Open add space factor	0.5	0.5

Keep **all symmetry planes set to none**. The first boundary definition in the original history had a magnetic Y-symmetry plane, but it was later removed: the offset lines are not mirror-symmetric about y = 0, so that plane would give wrong results.

About Zmax: with Zmax electric and no background spacing, CST puts a metal lid at the top of the bounding box (z = 10 mm), touching the top of the sphere. That turns the model into a shielded parallel-plate resonator. Use expanded open to replicate an open DR or DRA. Use electric only if you want an exact copy of the original results.

Step 8 — Define the waveguide ports

Put one waveguide port on the outer end face of each microstrip line.

1. Press **F** (Modeling → Picks → Pick Face) and click the end face of `solid4` at x = −25.
2. **Simulation → Waveguide Port**, then set:

Field	Value
Port number	1
Orientation	positive (from the picked face)
Position	Picks, Port on bound = on
Number of modes	1
Y extension (both sides)	`2.5*w_mic`
Z extension, lower	`t_sub` (reaches down to the ground at z = 0)
Z extension, upper	`2.5*w_mic`

3. Pick the end face of `solid4_1` at x = +25 and repeat with **Port number 2**, using the same extensions.

The resulting port is about 6 mm wide and 3 mm tall. That is roughly 6× the line width and 6× the substrate height, a usual size for microstrip ports. The port must reach the ground, otherwise the quasi-TEM mode is not captured. In the original history, the lower extension was first `t_sub+t_metal` and then corrected to `t_sub` ("modify port" steps).

Step 9 — Frequency range, solver and monitors

Sweep 9–14 GHz with the tetrahedral frequency-domain solver, and place field monitors at the resonances.

9a. Frequency range: Simulation → Frequency, with Fmin = `f_min` and Fmax = `f_max`.

9b. Solver: Simulation → Setup Solver (Frequency Domain). These are the settings that matter from the original model; leave everything else at its default.

Setting	Value
Method	Tetrahedral, General purpose
Order	Second
Adaptive mesh refinement	On (tetrahedral)
Stop criterion	All S-parameters, Δ = 0.01, 2 checks
Accuracy	1e-4
Stimulation	Port 1, mode 1 (enough for S11 and S21, since the structure is reciprocal)
Normalize to fixed impedance	On, 50 Ω
Frequency samples	Automatic, 1001 result samples
Mesh	High Frequency

9c. Field monitors: Simulation → Field Monitor. For each frequency, add an E-field, an H-field and a Power-flow monitor:

Frequency (GHz)	Likely mode
9.957	Magnetic dipole (TE_{101}-like)
10.34	Split or hybrid magnetic-dipole mode

Frequency (GHz)	Likely mode
13.45	Electric dipole (TM_{101}-like)
13.6	Electric dipole, companion peak

On a first run you won't know the exact peaks yet. Run once with no monitors, read the S21 peaks, then add monitors at those frequencies and re-run.

9d. Radiating-DRA extras (skip these for a pure two-port copy):

- Add **Far-field / RCS** monitors at the same frequencies (Simulation → Field Monitor → Farfield).
- Make sure Zmax is expanded open (Step 7).

Step 10 — Run and post-process

Run the solver, read the resonances off S21, confirm each mode from its field plot, then sweep the knobs.

1. **Check the mesh first: Home → Mesh View**. The sphere surface should look smooth, and adaptive refinement will add cells where needed.
2. **Run: Simulation → Start Simulation**. Watch the adaptive passes converge in the message window ($\Delta S < 0.01$).
3. **S-parameters:** open **1D Results → S-Parameters** and show S1,1 and S2,1 in dB. Use **Curve markers** (right-click → Add marker) on each S21 peak and S11 dip.
4. **Q factor:** for each S21 peak at f_0, measure the −3 dB bandwidth Δf, then:

```
Q_L = \frac{f_0}{\Delta f_{3dB}}, \qquad Q_u = \frac{Q_L}{1 - |S_{21}(f_0)|}
```

Here |S21| is the linear magnitude at the peak (10^(dB/20)). The Qu formula applies to a transmission-type resonator like this one.

5. **Identify each mode:** open **2D/3D Results → E-Field / H-Field** at each monitor and use a cut plane through the sphere centre. A magnetic-dipole mode shows the E-field circulating inside the sphere, with H along one axis. An electric-dipole mode shows the E-field running straight through the sphere.
6. **Parametric study: Simulation → Par. Sweep**, then sweep:
   - `r_sphere` from 4.5 to 5.5 in steps of 0.25 (moves the frequency)
   - `s_mic` from 0.5 to 2 (changes coupling and Q)

- `ss_mic` from 0 to 4 (changes which modes are excited)

7. **Export for your report (or for Claude to analyse):**
   - Right-click S-Parameters → **Export → Touchstone** (`.s2p`)
   - Any curve → **Export → Plot Data (ASCII)**
   - Screenshots of the field plots
   - **File → Export → STEP**, if you need the geometry

A successful replication has an S21 peak near 9.96 GHz and another near 13.5 GHz, matching the monitor frequencies of the original model.

Pitfalls and checklist

Most failed replications come from the boundaries, the ports or the overlap handling, not from the geometry.

Pitfall	Symptom	Fix
Zmax left electric	Resonances shifted, no radiation	Set Zmax to expanded open for an open DR or DRA
Symmetry plane left on	Wrong or missing peaks	All symmetry planes set to none (the lines are offset)
Port not reaching ground	Bad port mode, noisy S11	Lower Z extension = `t_sub`
Sphere–substrate overlap not removed	CST overlap warning, material ambiguity	Subtract the `w_inset` cylinder from `solid1`
Hard-coded numbers in dialogs	Parameter sweep doesn't move shapes	Type parameter names only
1 mm line vs 50 Ω port	Small ripple in S11	Accept it, or use `w_mic` = 1.25
Sphere drawn with segments > 0	Faceted sphere, frequency error	Segments = 0
Monitors at guessed frequencies	Field plots off resonance	Run once, read the peaks, then set the monitors

Final checklist

- [ ] Units mm / GHz, solver HF Frequency Domain

- [ ] 11 parameters defined; `w_inset` = 2.179
- [ ] 3 materials loaded from the library
- [ ] Substrate 50 × 30 × 0.5 with a Ø 4.36 mm hole
- [ ] Sphere r = 5 at (0, 0, 5), touching z = 0
- [ ] Two 1 mm lines, ends at x = ±3.18, y offsets −3 and +3
- [ ] Zmin electric; Xmin, Xmax, Ymin and Ymax open; symmetry none
- [ ] Ports 1 and 2 on the line ends, extended down to ground
- [ ] 9–14 GHz, tetrahedral, 2nd order, adaptive mesh
- [ ] S21 peaks near 9.96 and 13.5 GHz confirmed

Appendix — Rebuild with a macro

To skip the clicking, define the parameters with macro A, then paste block B into the History List. This is a trimmed copy of your original `model.mod`, with material fields reduced to the ones that affect the EM result.

A. Parameters: Home → Macros → New VBA Macro, paste, then Run.

```
Sub Main
    StoreParameter "l_sub", "50"
    StoreParameter "w_sub", "30"
    StoreParameter "t_sub", "0.5"
    StoreParameter "r_sphere", "5"
    StoreParameter "w_inset", "(r_sphere^2-(r_sphere-t_sub)^2)^0.5"
    StoreParameter "s_mic", "1"
    StoreParameter "w_mic", "1"
    StoreParameter "ss_mic", "3"
    StoreParameter "t_metal", "0.018"
    StoreParameter "f_min", "9"
    StoreParameter "f_max", "14"
    RebuildOnParametricChange False, True
End Sub
```

B. Model history: Home → History List → Edit, paste at the end, then **Update**. You can also rebuild it step by step with Steps 1–10 and compare against this text.

```
'@ solver and units
ChangeSolverType "HF Frequency Domain"
With Units
     .Geometry "mm"
     .Frequency "GHz"
     .Time "s"
End With

'@ materials
With Material
     .Reset
     .Name "Rogers RO3003 (loss free)"
     .FrqType "all"
     .Type "Normal"
     .SetMaterialUnit "GHz", "mm"
     .Epsilon "3"
     .Mu "1.0"
     .TanD "0.0"
     .Colour "0.75", "0.95", "0.85"
     .Create
End With
With Material
     .Reset
     .Name "Alumina (99.5%) (lossy)"
     .FrqType "all"
     .Type "Normal"
     .SetMaterialUnit "MHz", "mm"
     .Epsilon "9.9"
     .Mu "1.0"
     .TanD "0.0001"
     .TanDFreq "1.0"
     .TanDGiven "True"
     .TanDModel "ConstTanD"
     .Colour "0.94", "0.82", "0.76"
     .Create
End With
With Material
     .Reset
     .Name "Copper (annealed)"
     .FrqType "all"
     .Type "Lossy metal"
     .SetMaterialUnit "GHz", "mm"
     .Mu "1.0"
```

```
     .Kappa "5.8e+007"
     .Colour "1", "1", "0"
     .Create
End With

'@ substrate
Component.New "component1"
With Brick
     .Reset
     .Name "solid1"
     .Component "component1"
     .Material "Rogers RO3003 (loss free)"
     .Xrange "-l_sub/2", "l_sub/2"
     .Yrange "-w_sub/2", "w_sub/2"
     .Zrange "0", "t_sub"
     .Create
End With

'@ sphere
With Sphere
     .Reset
     .Name "solid2"
     .Component "component1"
     .Material "Alumina (99.5%) (lossy)"
     .Axis "z"
     .CenterRadius "r_sphere"
     .TopRadius "0"
     .BottomRadius "0"
     .Center "0", "0", "r_sphere"
     .Segments "0"
     .Create
End With

'@ seating hole
With Cylinder
     .Reset
     .Name "solid3"
     .Component "component1"
     .Material "Vacuum"
     .OuterRadius "w_inset"
     .InnerRadius "0.0"
     .Axis "z"
     .Zrange "0", "t_sub"
```

```
     .Xcenter "0"
     .Ycenter "0"
     .Segments "0"
     .Create
End With
Solid.Subtract "component1:solid1", "component1:solid3"

'@ feed line 1 + two mirrors
With Brick
     .Reset
     .Name "solid4"
     .Component "component1"
     .Material "Copper (annealed)"
     .Xrange "-w_inset-s_mic", "-l_sub/2"
     .Yrange "-w_mic/2-ss_mic", "w_mic/2-ss_mic"
     .Zrange "t_sub", "t_metal+t_sub"
     .Create
End With
With Transform
     .Reset
     .Name "component1:solid4"
     .Origin "Free"
     .Center "0", "0", "0"
     .PlaneNormal "0", "1", "0"
     .MultipleObjects "True"
     .GroupObjects "False"
     .Repetitions "1"
     .MultipleSelection "False"
     .Transform "Shape", "Mirror"
End With
With Transform
     .Reset
     .Name "component1:solid4_1"
     .Origin "Free"
     .Center "0", "0", "0"
     .PlaneNormal "1", "0", "0"
     .MultipleObjects "False"
     .GroupObjects "False"
     .Repetitions "1"
     .MultipleSelection "False"
     .Transform "Shape", "Mirror"
End With
```

```
'@ boundaries  (original: Zmax electric; use "expanded open" for a radiating
DRA)
With Boundary
     .Xmin "expanded open"
     .Xmax "expanded open"
     .Ymin "expanded open"
     .Ymax "expanded open"
     .Zmin "electric"
     .Zmax "electric"
     .Xsymmetry "none"
     .Ysymmetry "none"
     .Zsymmetry "none"
     .OpenAddSpaceFactor "0.5"
End With

'@ ports
Pick.PickFaceFromId "component1:solid4", "4"
With Port
     .Reset
     .PortNumber "1"
     .NumberOfModes "1"
     .Coordinates "Picks"
     .Orientation "positive"
     .PortOnBound "True"
     .ClipPickedPortToBound "False"
     .XrangeAdd "0.0", "0.0"
     .YrangeAdd "2.5*w_mic", "2.5*w_mic"
     .ZrangeAdd "t_sub", "2.5*w_mic"
     .Create
End With
Pick.PickFaceFromId "component1:solid4_1", "4"
With Port
     .Reset
     .PortNumber "2"
     .NumberOfModes "1"
     .Coordinates "Picks"
     .Orientation "positive"
     .PortOnBound "True"
     .ClipPickedPortToBound "False"
     .XrangeAdd "0.0", "0.0"
     .YrangeAdd "2.5*w_mic", "2.5*w_mic"
     .ZrangeAdd "t_sub", "2.5*w_mic"
     .Create
```

```
End With

'@ frequency + solver
Solver.FrequencyRange "f_min", "f_max"
Mesh.SetCreator "High Frequency"
With FDSolver
     .Reset
     .SetMethod "Tetrahedral", "General purpose"
     .OrderTet "Second"
     .Stimulation "1", "1"
     .AutoNormImpedance "False"
     .NormingImpedance "50"
     .AccuracyTet "1e-4"
     .MeshAdaptionTet "True"
     .SetNumberOfResultDataSamples "1001"
End With

'@ field monitors (repeat for 10.34, 13.45, 13.6)
With Monitor
     .Reset
     .Name "e-field (f=9.957)"
     .Dimension "Volume"
     .Domain "Frequency"
     .FieldType "Efield"
     .MonitorValue "9.957"
     .Create
End With
With Monitor
     .Reset
     .Name "h-field (f=9.957)"
     .Dimension "Volume"
     .Domain "Frequency"
     .FieldType "Hfield"
     .MonitorValue "9.957"
     .Create
End With
```

The face ID "4" in `Pick.PickFaceFromId` is the outer end face of each line in the original model. If a port lands on the wrong face, delete it and pick the face by hand (Step 8).